

CO2xNxPhrag: Porewater Ammonium

May 2026 Samples

2026-08-07

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##Run Information

```
cat("Run Information: Brenden Blakley") #lets you know what section you're in
```

Run Information: Brenden Blakley

```
library(here)
```

here() starts at C:/Users/blakleybd/OneDrive - Smithsonian Institution/Documents/GitHub/GCREW

```
#set the run date & user name
```

```
sample_year <- "2026"
```

```
sample_month <- "May"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month
```

```
NH3_files <- c("Raw Data/20260608_2M_Phragrr_May_Nh4.csv",  
              "Raw Data/20260608_Phrag_May_NH4.csv")
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
file_path <- here("Porewater QAQC Logs", "GCREW_NH4_QAQC_Log.csv")
```

```
file_path_QAQC <- here("Porewater QAQC Logs", "GCREW_SEAL_Check_QAQC_Log.csv")
```

```
final_path <- "Processed Data/Phrag_Porewater_NH4_202605.csv"
```

```
#record any notes about the run or anything other info here (Leave empty otherwise):
```

```
run_notes <- "Two Separate runs done on 20260608, one PE check passes one fails"
```

```
cat(run_notes)
```

Two Separate runs done on 20260608, one PE check passes one fails

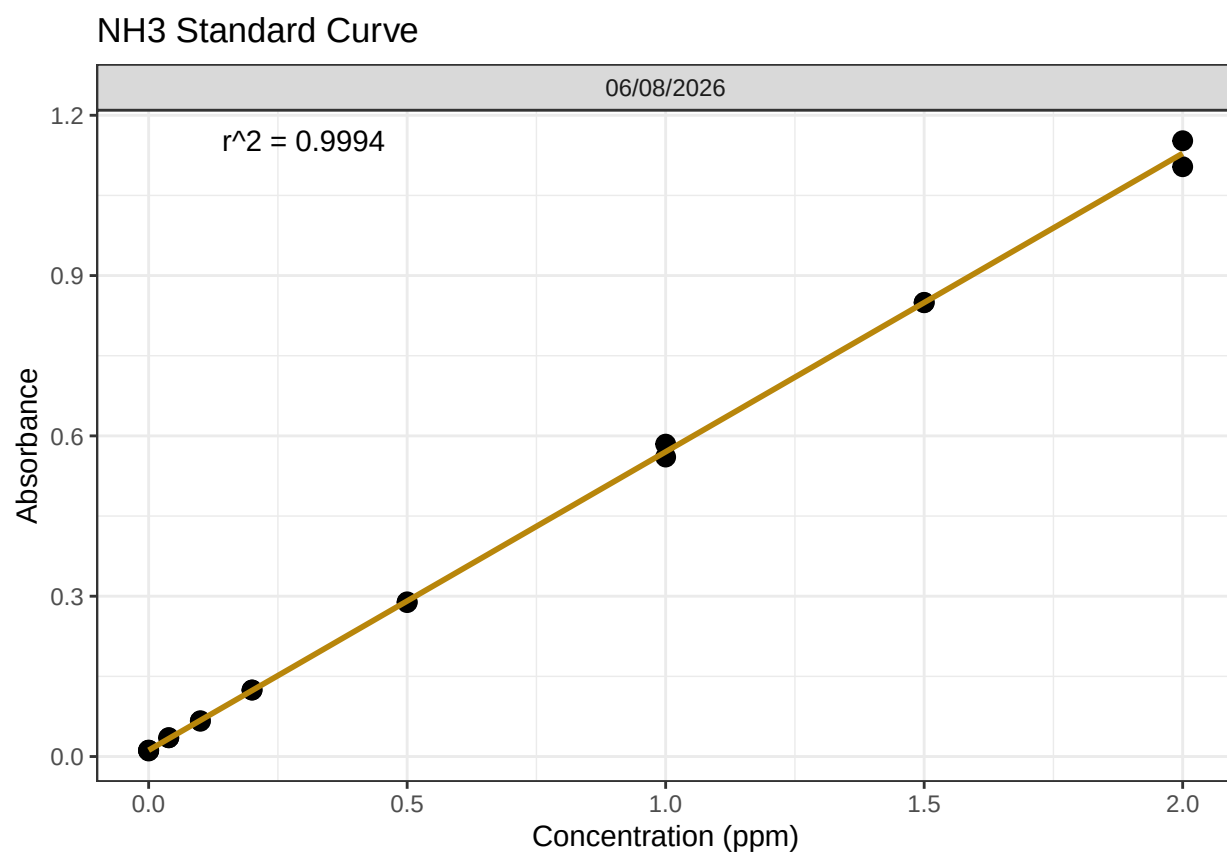
##Setup

##Read in metadata and create similar sample IDs for matching to samples

```
##Import Data & Clean
```

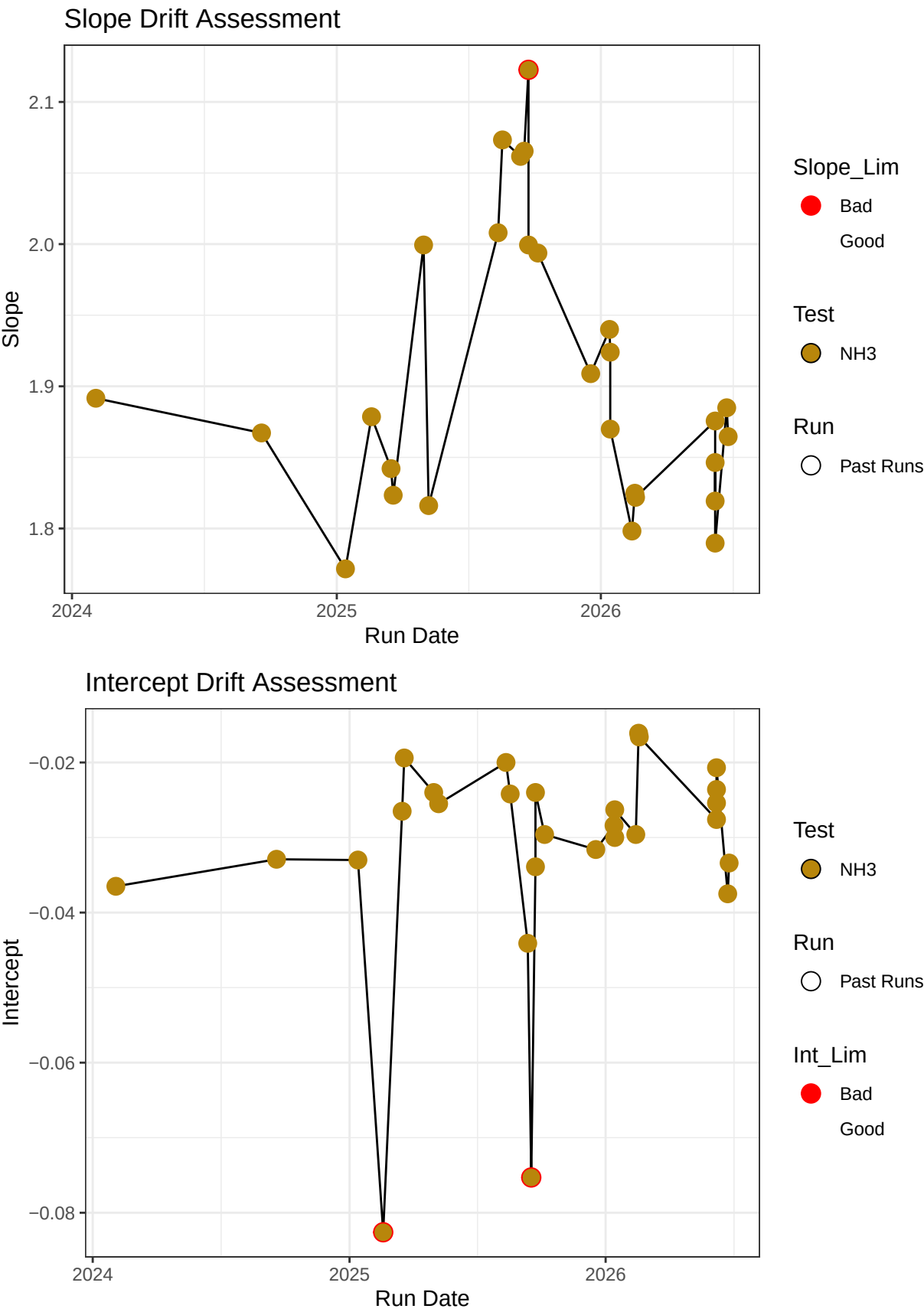
0.1 Assessing standard Curves

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Test	Run_Date	R2_Flag	R2	Slope
Ammonia 2	06/08/2026	R2_Pass	0.999414	1.789672

0.2 Update Standard Log



0.3 Dilution Corrections - ensure the latest dilution is kept

```
## [1] "No Reruns"
```

```
## [1] "No Dilutions"
```

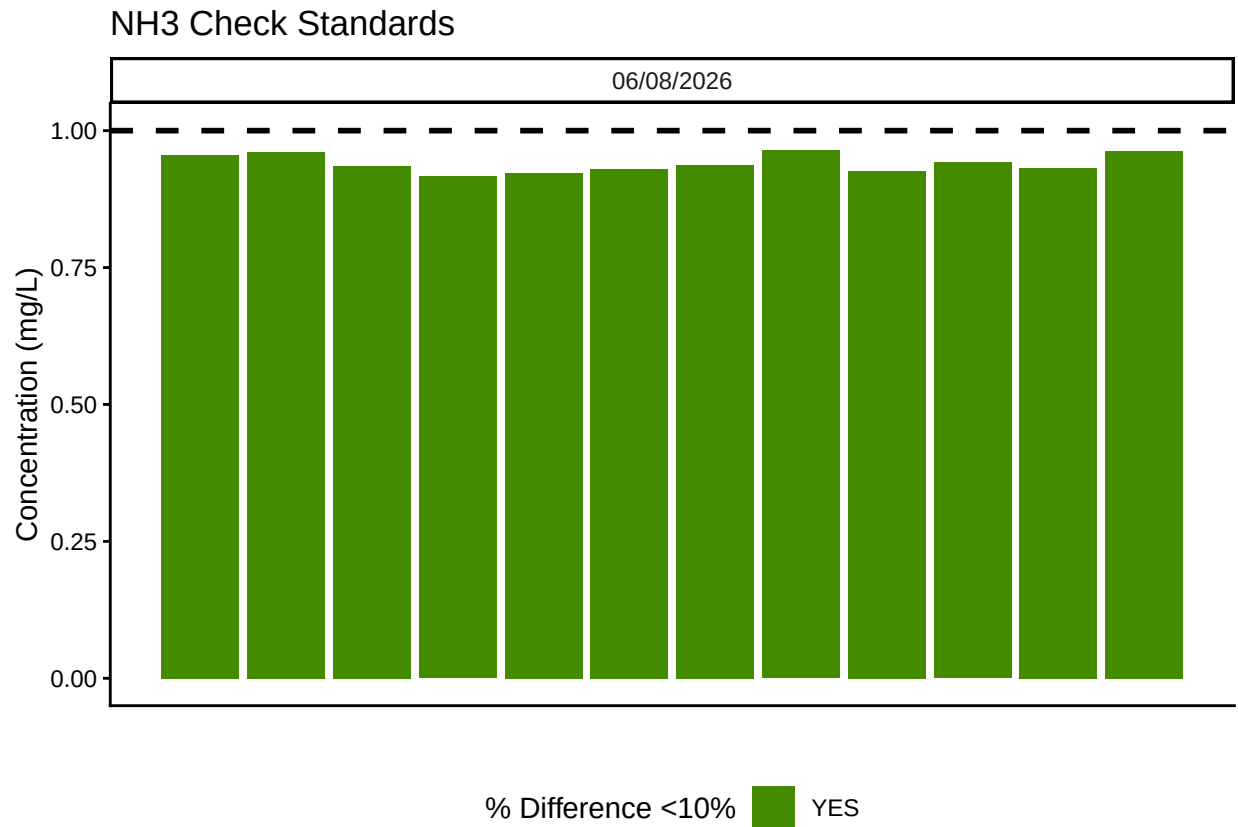
```
## [1] "No Naming Issues Detected"
```

0.4 Performance Check

Test	Run_Date	PE_Flag	PE_Conc	PE_Target_Conc
Ammonia 2	06/08/2026	Performance Check NOT Within 25% - REASSESS	1.297496	1.034
Ammonia 2	06/08/2026	Performance Check NOT Within 25% - REASSESS	1.297496	1.034

0.5 Analyze the Check Standards

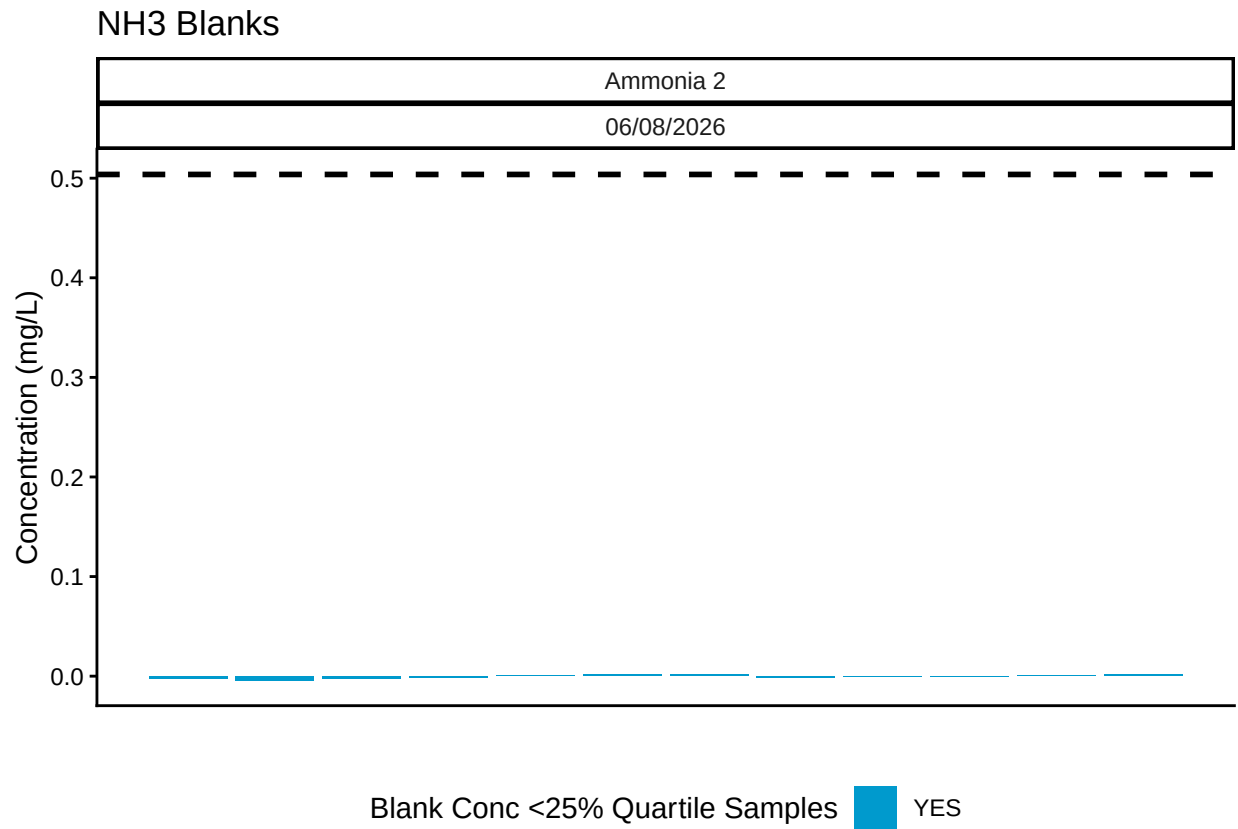
Test	Run_Date	RSV_Flag	RSV	RSV_Cutoff
Ammonia 2	06/08/2026	RSV WITHIN RANGE - PROCEED	0.0176254	0.25



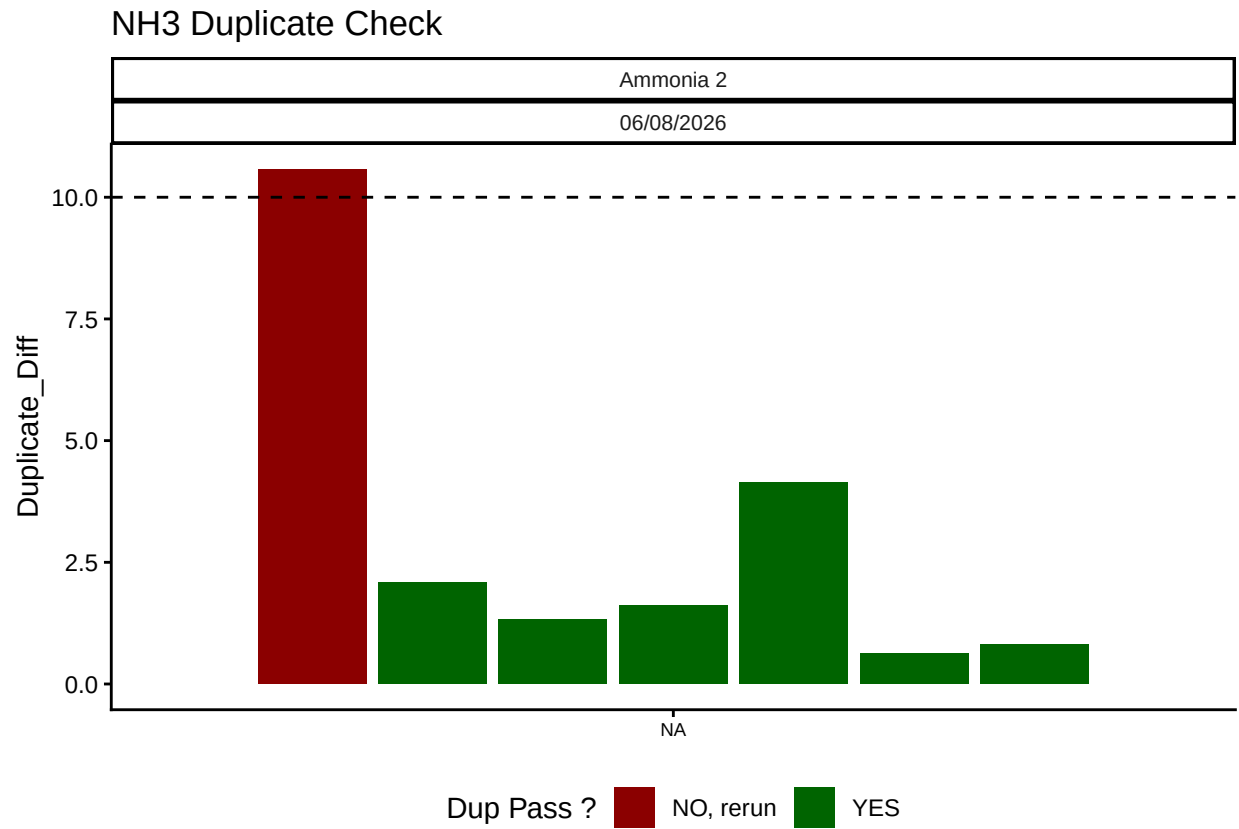
Test	Run_Date	CHK_Flag
Ammonia 2	06/08/2026	>60% of Checks Pass - PROCEED

0.6 Analyze Blanks

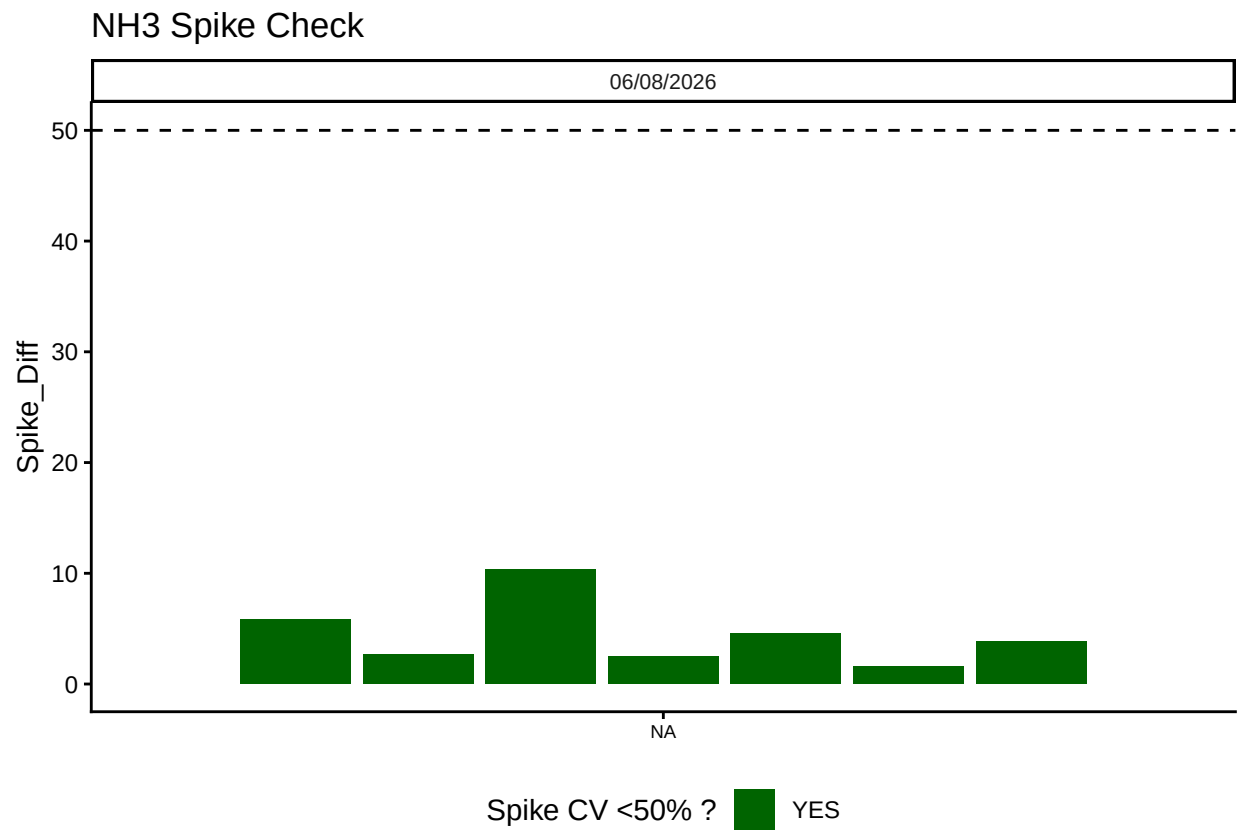
Test	Run_Date	BLK_Pct_Flag	Mean_Blkc_Conc	Quantile_25
Ammonia 2	06/08/2026	>60% of Blanks Pass - PROCEED	-0.0005872	0.503759



0.7 Analyze Duplicates



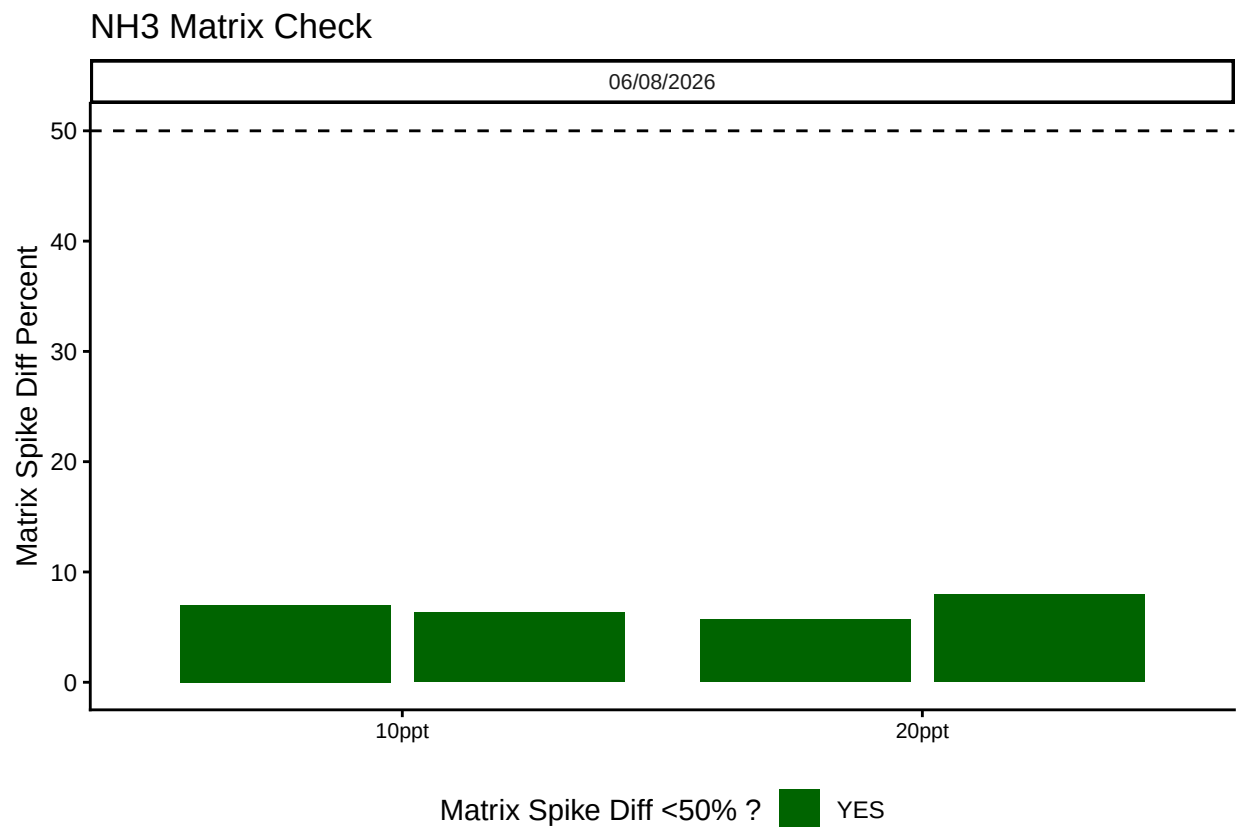
0.8 Spikes



Test	Run_Date	Spike_Flags
Ammonia 2	06/08/2026	>60% of Spikes have a Diff <50% - PROCEED

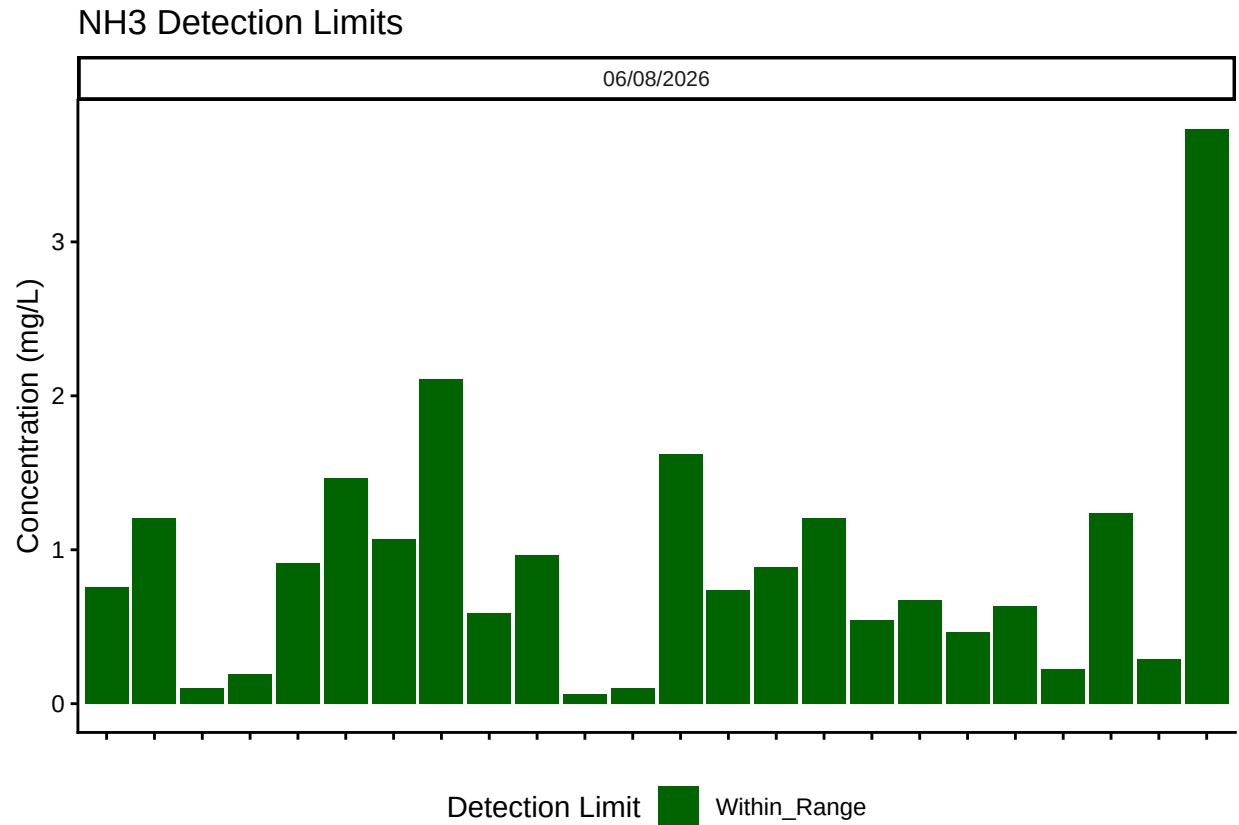
0.9 Matrix Effects

```
## Warning in left_join(Matrix, Spike_ready, by = join_by(Date_Num)): Detected an unexpected many-to-many
## i Row 1 of 'x' matches multiple rows in 'y'.
## i Row 25 of 'y' matches multiple rows in 'x'.
## i If a many-to-many relationship is expected, set 'relationship =
##   "many-to-many"' to silence this warning.
```



Test	Run_Date	Matrix_Flags
Ammonia 2	06/08/2026	Matrix Has Diff <50% - PROCEED

```
##Sample Flagging - Within range of standard curve
```

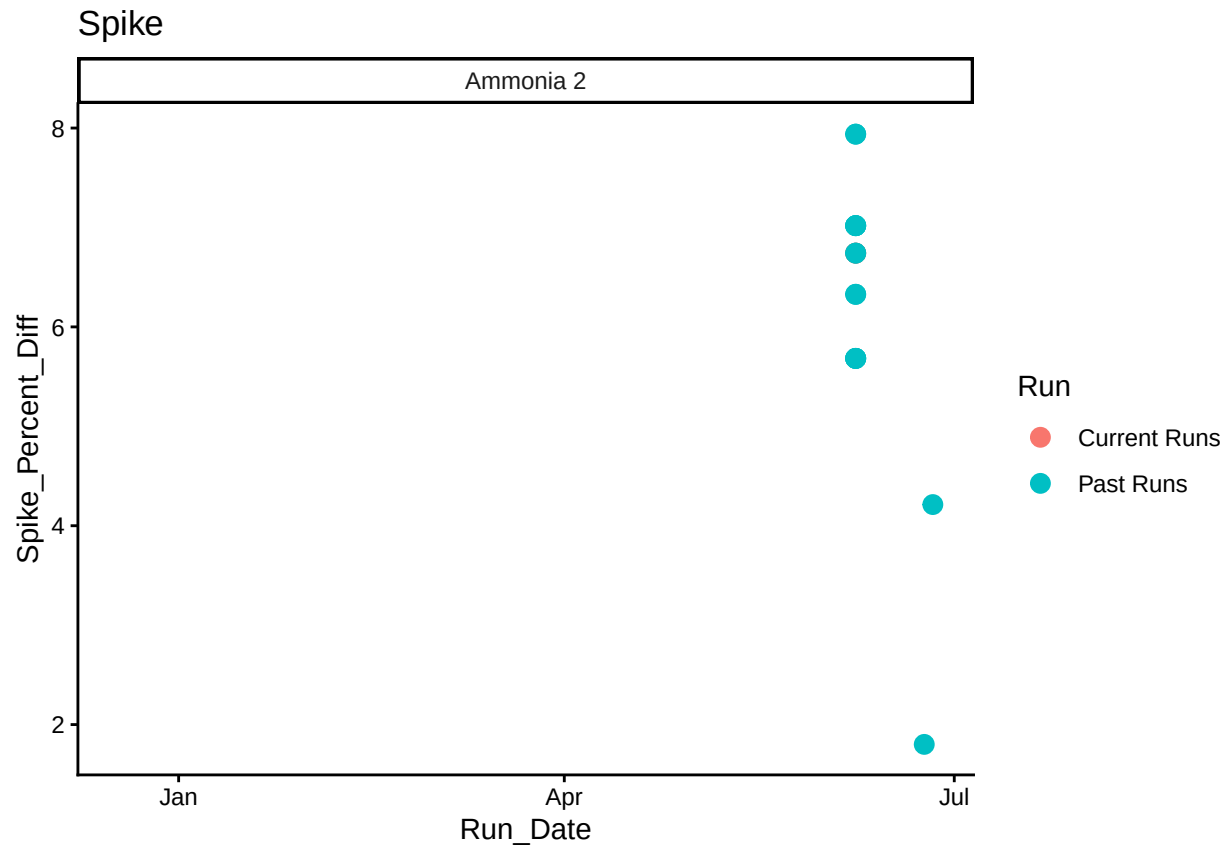


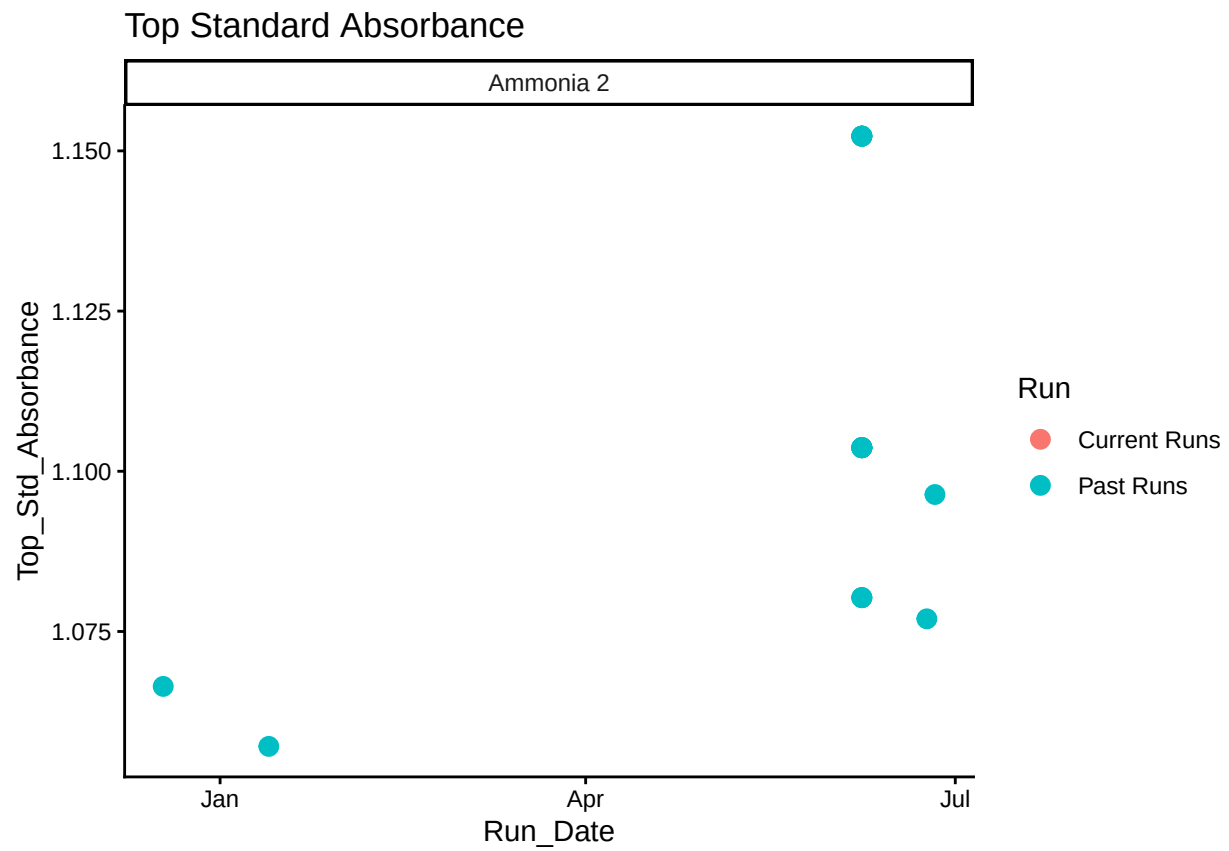
```
##Add QAQC Flags
```

0.10 Create Log for QAQC

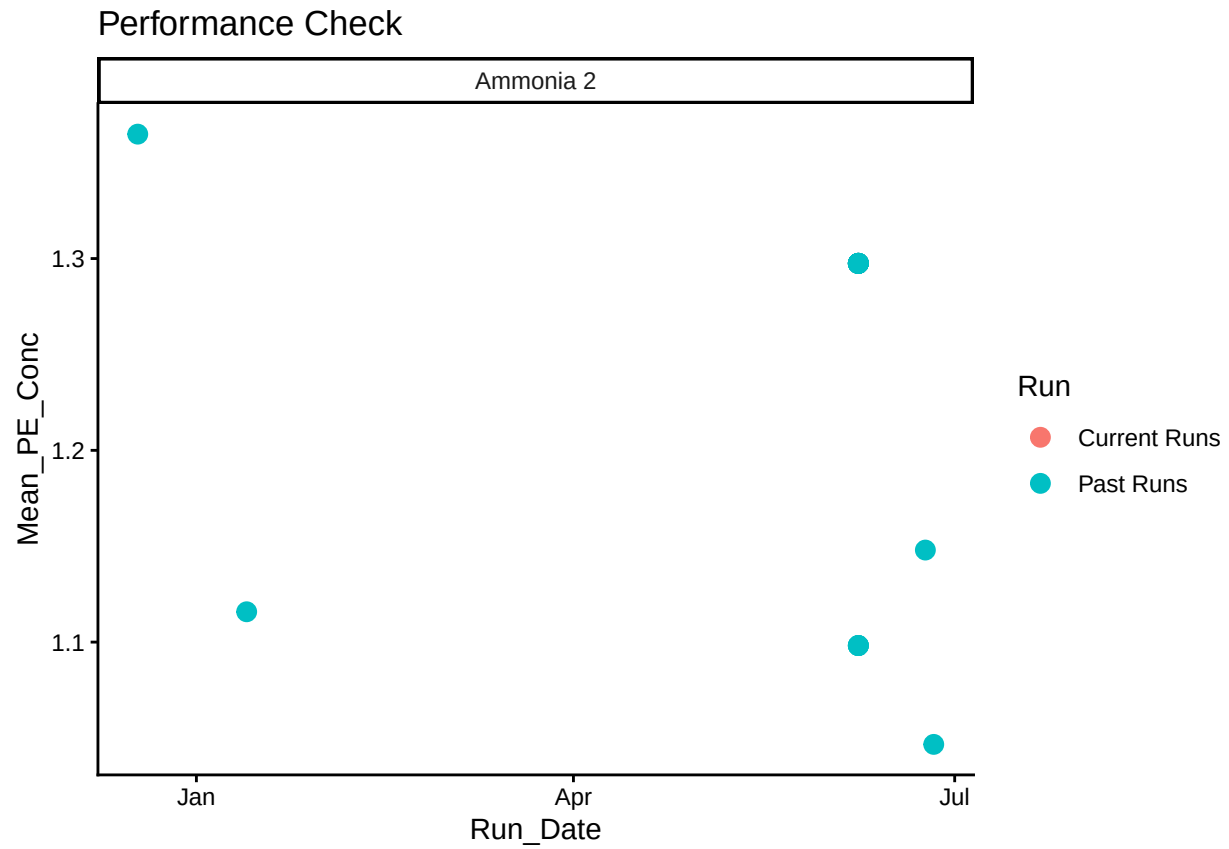
```
## Warning in left_join(Top_Standards, PE_Check, by = join_by(Test, Run_Date)): Detected an unexpected relationship between
## i Row 1 of 'x' matches multiple rows in 'y'.
## i Row 1 of 'y' matches multiple rows in 'x'.
## i If a many-to-many relationship is expected, set 'relationship =
##   "many-to-many"' to silence this warning.

## Warning: Removed 3 rows containing missing values or values outside the scale range
## ('geom_point()').
```





```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



0.11 Merge samples with Metadata & check all samples are present

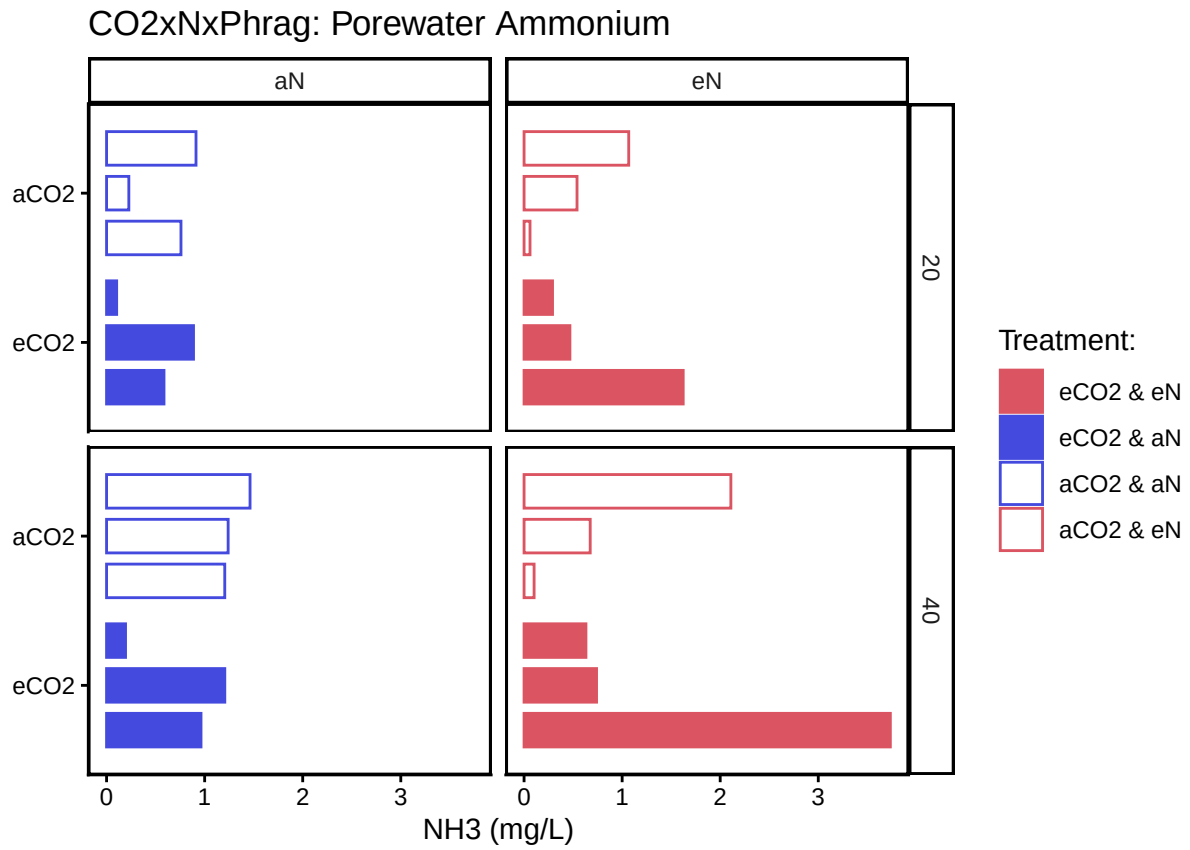
```
## Merge Samples with Metadata
```

```
## All sample IDs are present in data
```

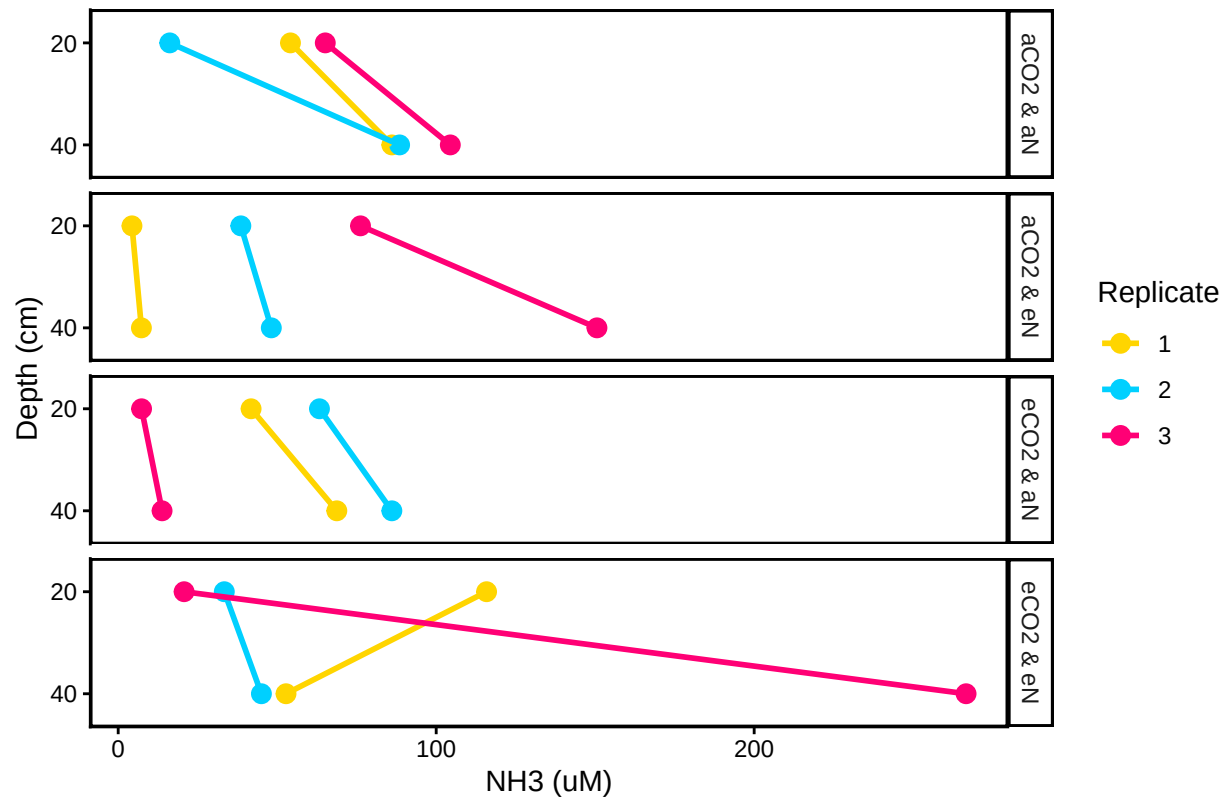

0.12 Organize Data

0.13 Visualize Data

Visualize Data



CO₂xNxPhrag: Porewater NH₃ by Replicate



##Export Data

#end